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PUBLICATIONS

- H. Kumar, **A. Konkar**. *Simple Transformer with Single Leaky Neuron for Event Vision*. **Feb 2025**
Proceedings of the Winter Conference on Applications of Computer Vision (WACV) Workshops, 2025, pp. 928-934
- A. Konkar**, X. Qu. *A Review of Transformer-Based and Hybrid Deep Learning Approaches for EEG Analysis*. **Jun 2025**
International Conference on Human-Computer Interaction (HCI International) 2025.

EDUCATION

- Master of Science**, Computer Science **May 2025**
George Washington University **GPA: 3.71/4.0**
Thesis: Enhancing EEG-Based Gaze Prediction with Transformers on EEGEyeNet 
Relevant Courses: Computational Linear Algebra, Computer Vision, Machine Learning, Neural Networks & Deep Learning.
- Bachelor of Engineering**, Information Technology **Oct 2020**
University of Mumbai

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, C#, MATLAB, R, SQL, JavaScript, TypeScript
Frameworks: PyTorch, Tensorflow, Keras, Spring Boot, ASP.NET Core, Angular, React, Next.js, Selenium
ML Algorithms: PCA, SVC, K-means, Maximum Likelihood Estimation, Bayesian Estimation, Gaussian Mixture Models, Expectation-Maximization, Gradient Descent (SGD, Adam), SVM
DL Architectures: MLP, CNN, RNN, LSTM, Self-Attention, Transformers, Attention-Based Fusion, VAE, GAN, LLM
Computer Vision: Optical Flow, Visual Odometry, SfM, SLAM, NeRF, 3D Gaussian Splatting, Diffusion Models, Multiple View Geometry, Sensor Fusion, Event-based Vision

EXPERIENCE

- Research Assistant – Computer Vision** **Oct 2025 – Present**
GW Vision Lab *Washington, DC*
- Creating a dataset of event camera recordings of various objects responding to sound stimuli, where audio not recorded.
 - Since event cameras can capture tiny, rapid brightness changes, sound becomes visible in the form of motion as it physically vibrates objects. Checkout this post to view more information: [See Motion from Sound with Event Cameras](#).
 - Investigating methods to reconstruct acoustic signals from event-based visual input. Research Advisor: Dr. Robert Pless.
- Software Engineer** **Mar 2025 – Present**
National Collegiate Table Tennis Association (NCTTA) *Washington, DC*
- Developing and shipping production-ready features using ASP.NET Core for the NCTTA web application serving **20,000+** users (see live app: <https://nctta.app/>). This app manages all U.S. collegiate-level table tennis tournaments.
 - Designed and implemented an end-to-end Contact Us support system with database-backed request tracking, admin resolution workflows, and IP-based rate limiting (10 requests / 10 minutes per IP) to prevent abuse while ensuring graceful user feedback.
 - Improved service performance and scalability by **20%** through SQL query optimization, indexing strategies, and streamlining request-response data flows to reduce backend latency and server load.
- Research Assistant – Data Science** **Nov 2023 – Oct 2025**
GW Institute of Public Policy *Washington, DC*
- Performed end-to-end data engineering, statistical analysis, and causal modeling to evaluate the impacts and external validity of large-scale career pathways programs. Tech Stack: R, Python, SQL.
 - Developed data preprocessing pipelines by performing attribute selection, appropriate inner joins, and schema alignment to transform raw, multi-source datasets into analysis-ready tables.
 - Implemented hierarchical causal models to estimate group-specific treatment effects, using LASSO and Empirical Bayes regularization to improve generalization and robustness.
 - Evaluated model generalization using out-of-sample projections and uncertainty estimates to assess how well impacts transport across sites and settings.
 - Outcome:** Provided a framework for site-specific impact estimation with quantified uncertainty, aimed at informing adoption decisions beyond average effects.

- Developed **Spring Boot microservices** and implemented **Selenium-based test automation** for an internal Capital Markets platform for our client, *Citi Bank*, ensuring compliance with corporate QA standards.
- Saved **8 hours** of manual testing effort **per week** by automating complex end-to-end test scenarios.
- Collaborated with cross-functional teams to translate functional specifications into modular, maintainable software components. Participated in code reviews and iterative feature delivery. Followed Agile methodologies and TDD.
- Optimized SQL queries and improved API efficiency, contributing to a **25%** reduction in data retrieval time.
- Deployed Spring Boot microservices on AWS EC2 via OpenShift, enhancing scalability and performance.

TEACHING EXPERIENCE

Teaching Assistant – CSCI 1011. Introduction to Software Development, GW	May 2025 – Aug 2025
Teaching Assistant – CSCI 1112. Algorithms and Data Structures, GW	May 2025 – Aug 2025
Teaching Assistant – CSCI 2113. Software Engineering, GW	May 2025 – Aug 2025
Student Tutor – <i>GW Athletics</i>	Sep 2023 – May 2025

- MATH 1221. Calculus with Precalculus II.
- MATH 1232. Single-Variable Calculus II.
- MATH 3125. Linear Algebra II.
- CSCI 1011. Introduction to Programming with Java.
- CSCI 1112. Algorithms and Data Structures.

SELECTED PROJECTS

FlavorSignal – Serpapi, Ollama(Phi-3 Mini), Next.js, Python [GitHub](#)

- **Outcome:** Given your location and favorite dish as input, FlavorSignal fetches all the nearby restaurants and tells you whether that dish is worth ordering by cutting through noisy reviews and summarizing opinions specific to your dish.
- **Approach:** FlavorSignal filters reviews semantically at the dish level, using query expansion and embedding-based retrieval to isolate only text relevant to the requested item.
- The filtered, item-specific reviews are summarized to provide a concise assessment of whether a dish is worth ordering, including sentiment trends and common complaints.

From-Scratch Implementation of a Low-Level Image Classification Network – Python [\[Report\]](#) [GitHub](#)

- Developed a complete neural network training pipeline in Python (without using deep learning libraries).
- Implemented manual forward pass, backpropagation, gradient updates, and weight initialization.
- Built custom image preprocessing modules (posterization, enhancement, feature extraction) to improve data quality and model performance.

Forecasting Hourly Electricity Demand and Assessing Grid Resilience – Python [\[Report\]](#)

- Fine-tuned machine learning models (LSTM and Prophet) to predict hourly electricity demand from national grid data.
- Evaluated how major disruptions (like storms or accidents) influence grid performance, turning data-driven insights into recommendations for stronger energy systems.

Landmark Recognition – Python, Streamlit [GitHub](#)

- Developed a landmark recognition web application that predicts landmarks from images, retrieves their full address with latitude/longitude, and visualizes them on an interactive map for exploration.
- Leveraged a pretrained tensorflow-hub model, trained on the Google Landmarks Dataset V2.

Real-Time Person Detection & Tracking – Python [GitHub](#)

- Developed a real-time person detection and tracking pipeline using YOLOv8x for detection and DeepSORT for tracking.
- Evaluated multiple tracking methods (IOU, SORT, DeepSORT). Tested on an NVIDIA RTX 3070.

REFERENCES

Robert Pless

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George Washington University
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Robert Olsen

Research Professor at George Washington Institute of Public Policy
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